



Oklahoma State Standards Alignment

For Grades K-8

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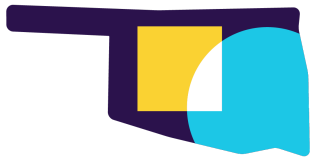
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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to Oklahoma's Academic Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to Oklahoma's Academic Standards for Mathematics to fully support Oklahoma's students and teachers. The tables on the following pages, organized by Oklahoma Academic Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Counting & Cardinality		
K.N.1.1	Count aloud forward in sequence to 100 by 1s and 10s	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.N.1.2	Recognize that a number can be used to represent how many objects are in a set up to 10	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.N.1.3	Use ordinal numbers to represent the position of an object in a sequence up to 10.	Mission 6, Topic A
K.N.1.4	Recognize without counting (subitize) the quantity of a small group of objects in organized and random arrangements up to 10.	Mission 4, Topic A
K.N.1.5	Count forward, with and without objects, from any given number up to 20.	Mission 1, Topic G Mission 5, Topic A Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E
K.N.1.6	Read, write, discuss, and represent whole numbers from 0 to at least 20. Representations may include numerals, pictures, real-object and pictographs, spoken words, and manipulatives.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
K.N.1.7	Find a number that is 1 more or 1 less than a given number up to 10.	Mission 1, Topic G Mission 1, Topic H

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.N.1.8	Compare and order whole numbers from 0 to 10 with and without objects, using the vocabulary "more than," "less than," or "equal to."	Mission 3, Topic B Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G Mission 3, Topic H Mission 5, Topic E
K.N.2.1	Compose and decompose numbers up to 10 using objects and pictures.	Mission 1, Topic C Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 4, Topic G
K.N.3.1	Distribute a set of objects into at least two smaller equal sets.	Grade 2, Mission 6, Topic A
K.N.4.1	Identify pennies, nickels, dimes, and quarters by name.	Grade 1, Mission 6, Topic E
Algebraic Reasoning & Algebra		
K.A.1.1	Sort and group up to 10 objects into a set based upon characteristics such as color, size, and shape. Explain verbally what the objects have in common.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C
K.A.1.2	Recognize, duplicate, complete, and extend repeating, increasing, and decreasing patterns in a variety of contexts (i.e., shape, color, size, objects, sounds, movement).	Mission 1, Topic H Mission 4, Topic B Mission 4, Topic H Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
Geometry & Measurement		
K.GM.1.1	Recognize squares, circles, triangles, and rectangles.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 6, Topic A
K.GM.1.2	Sort two-dimensional objects using characteristics such as shape and size.	Mission 2, Topic A Mission 2, Topic C
K.GM.1.3	Identify attributes of two-dimensional shapes using informal and formal geometric language interchangeably, such as the number of corners/vertices and the number of sides/edges.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A Mission 6, Topic B
K.GM.1.4	Use smaller two-dimensional shapes to fill in the outline of a larger two-dimensional shape.	Mission 6, Topic B
K.GM.1.5	Compose larger, undefined shapes and structures using three-dimensional objects.	Mission 6, Topic B
K.GM.1.6	Use basic shapes and spatial reasoning to represent objects in the real world	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 6, Topic A
K.GM.2.1	Use words to compare objects according to length, size, weight, position, and location.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F Mission 3, Topic H
K.GM.2.2	Order up to 6 objects using measurable attributes, such as length and weight.	Grade 1, Mission 3, Topic A Grade 1, Mission 3, Topic B
K.GM.2.3	Identify more than one shared attribute between objects, and sort objects into sets	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C
K.GM.2.4	Compare the number of objects needed to fill two different containers.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
		Mission 1, Topic E Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C
K.GM.3.1	Develop an awareness of simple time concepts within daily life, using age-appropriate vocabulary (e.g., yesterday, today, tomorrow, morning, afternoon, night).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Data & Probability		
K.D.1.1	Collect and organize information about objects and events in the environment.	Grade 1, Mission 3, Topic D
K.D.1.2	Use categorical data to create real-object graphs and pictographs.	Grade 1, Mission 3, Topic D Grade 2, Mission 7, Topic A
K.D.1.3	Draw conclusions from real-object graphs and pictographs.	Grade 1, Mission 3, Topic D Grade 2, Mission 7, Topic A

1st Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Numbers & Operations		
1.N.1.1	Recognize numbers to 20 without counting (subitize) the quantity of structured arrangements	Grade K, Mission 5, Topic C
1.N.1.2	Use concrete representations to describe whole numbers between 10 and 100 in terms of tens and ones. Know that 10 is equivalent to 10 ones and 100 is equivalent to 10 tens.	Mission 2, Topic D Mission 4, Topic A Mission 4, Topic B Mission 4, Topic F Mission 6, Topic B
1.N.1.3	Read, write, discuss, and represent whole numbers up to 100. Representations may include numerals, words, addition and subtraction, pictures, tally marks, number lines, and manipulatives.	Mission 4, Topic A Mission 6, Topic B
1.N.1.4	Count forward, with objects, from any given number up to 100 by 1s, 2s, 5s and 10s.	Mission 4, Topic A Mission 6, Topic B
1.N.1.5	Count forward, without objects, by multiples of 1s, 2s, 5s, and 10s, up to 100.	Mission 4, Topic A Mission 6, Topic B
1.N.1.6	Find a number that is 10 more or 10 less than a given number up to 100.	Mission 2, Topic D Mission 4, Topic A Mission 6, Topic B
1.N.1.7	Compare and order whole numbers from 0 to 100.	Mission 4, Topic B Mission 6, Topic B
1.N.1.8	Use knowledge of number relationships to locate the position of a given whole number, up to 20, on an open number line.	Grade 2, Mission 7, Topic E
1.N.1.9	Use words such as “more than,” “less than,” and “equal to” to describe the relative value of numbers	Mission 4, Topic B
1.N.2.1	Represent and solve problems using addition and subtraction with sums and minuends of up to 10.	Mission 1, Topic A Mission 1, Topic B

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
		Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F
1.N.2.2	Determine if equations involving addition and subtraction are true.	Mission 1, Topic E Mission 2, Topic B Mission 2, Topic B
1.N.2.3	Demonstrate fluency with basic facts of addition and subtraction with sums and minuends of up to 10.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic F Mission 1, Topic I Mission 1, Topic J Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D
1.N.3.1	Partition a regular polygon using physical models and recognize when those parts are equal.	Mission 5, Topic C Mission 5, Topic D
1.N.3.2	Partition (fair share) sets of objects into two and three equal groups.	Grade 2, Mission 6, Topic A
1.N.4.1	Identify pennies, nickels, dimes, and quarters by name and value.	Mission 6, Topic E
1.N.4.2	Write a number with the cent symbol to describe the value of a coin.	Mission 6, Topic E
1.N.4.3	Determine the value of a collection of pennies, nickels, or dimes up to one dollar, counting by 1s, 5s, and 10s.	Mission 6, Topic E

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Algebraic Reasoning & Algebra		
1.A.1.1	Identify, create, complete, and extend repeating, increasing, and decreasing patterns in a variety of contexts (e.g., quantity, numbers, or shapes).	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic G Mission 1, Topic H Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 3, Topic C Mission 3, Topic D Mission 4, Topic E Mission 6, Topic A Mission 6, Topic F
Geometry & Measurement		
1.GM.1.1	Identify regular and irregular trapezoids and hexagons by pointing to the shape when given the name.	Mission 5, Topic A Grade K, Mission 2, Topic A
1.GM.1.2	Compose larger, defined shapes using smaller two-dimensional shapes.	Mission 5, Topic B
1.GM.1.3	Compose structures with three-dimensional shapes.	Mission 5, Topic B
1.GM.1.4	Recognize three-dimensional shapes such as cubes, cones, cylinders, pyramids, and spheres.	Grade K, Mission 2, Topic B
1.GM.2.1	Use nonstandard and standard measuring tools to measure the length of objects.	Grade K, Mission 2, Topic A Grade K, Mission 2, Topic B Grade K, Mission 2, Topic C Grade K, Mission 2, Topic D
1.GM.2.2	Illustrate that the length of an object is the number of same-size units of length that, when laid end-to-end with no gaps or overlaps, reach from one end of the object to the other.	Mission 3, Topic B Mission 3, Topic C

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.GM.2.3	Measure the same object/distance with units of two different lengths, and describe how and why the measurements differ.	Grade 2, Mission 2, Topic C
1.GM.2.4	Describe a length to the nearest whole unit using a number with standard and nonstandard units.	Grade 2, Mission 2, Topic B Grade 2, Mission 2, Topic D
1.GM.2.5	Use standard and nonstandard tools to identify volume/capacity. Compare and sort containers that hold more, less, or the same amount.	Grade 2, Mission 3, Topic C
1.GM.3.1	1 Tell time to the hour and half-hour (analog and digital).	Mission 5, Topic D
1.GM.3.2	Describe and measure calendar time by days, weeks, months, and years.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Data & Probability		
1.D.1.1	Collect, sort, and organize data in up to three categories using representations (e.g., tally marks, tables, Venn diagrams).	Mission 3, Topic D
1.D.1.2	Use data to create pictographs and bar graphs that demonstrate one-to-one correspondence.	Mission 3, Topic D
1.D.1.3	Draw conclusions from pictographs and bar graphs.	Mission 3, Topic D

2nd Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Numbers & Operations		
2.N.1.1	Read, write, discuss, and represent whole numbers up to 1,000. Representations should include, but are not limited to, numerals, words, pictures, tally marks, number lines, and manipulatives.	Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.N.1.2	Use knowledge of number relationships to locate the position of a given whole number, up to 100, on an open number line.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.N.1.3	Use place value to describe whole numbers between 10 and 1,000 in terms of hundreds, tens, and ones, including written, standard, and expanded forms. Know that 10 is equivalent to 10 ones and 100 is equivalent to 10 tens.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
2.N.1.4	Find 10 more or 10 less than a given three-digit number. Find 100 more or 100 less than a given three-digit number.	Mission 3, Topic G Mission 5, Topic A Mission 5, Topic D
2.N.1.5	Use objects to determine whether a number is even or odd.	Mission 6, Topic D
2.N.1.6	Use place value understanding to round numbers to the nearest ten and nearest hundred (up to 1,000). Recognize when to round in real-world situations.	Grade 3, Mission 2, Topic C
2.N.1.7	Use place value to compare and order whole numbers up to 1,000 using comparative language, numbers, and symbols (e.g., $425 > 276$, $73 < 107$, page 351 comes after page 350, 753 is between 700 and 800).	Mission 3, Topic E Mission 3, Topic F
2.N.2.1	Use the relationship between addition and subtraction to generate basic facts with sums and minuends of up to 20.	Mission 1, Topic A Mission 1, Topic B

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.N.2.2	Demonstrate fluency with basic facts of addition and subtraction with sums and minuends of up to 20.	Mission 1, Topic A Mission 1, Topic B
2.N.2.3	Estimate sums and differences up to 100.	Grade 3, Mission 2, Topic D
2.N.2.4	Use strategies and algorithms based on knowledge of place value and equality to add and subtract two-digit numbers.	Mission 1, Topic B Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 7, Topic B Mission 7, Topic E Mission 8, Topic D
2.N.2.5	Solve addition and subtraction problems involving whole numbers up to two digits.	Mission 1, Topic B Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic F
2.N.2.6	Use concrete models and structured arrangements, such as repeated addition, arrays, and ten frames to develop an understanding of multiplication.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
2.N.3.1	Identify the parts of a set and area that represent fractions for halves, thirds, and fourths.	Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D
2.N.3.2	Construct equal-sized portions through fair sharing (length, set, and area models for halves, thirds, and fourths).	Mission 8, Topic C Mission 8, Topic D
2.N.4.1	Determine the value of a collection of coins up to one dollar using the cent symbol.	Mission 7, Topic B
2.N.4.2	Use a combination of coins to represent a given amount of money up to one dollar.	Mission 7, Topic B
Algebraic Reasoning & Algebra		

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.A.1.1	Represent, create, describe, complete, and extend increasing and decreasing patterns with quantity and numbers in a variety of contexts.	Mission 1, Topic B Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D Mission 5, Topic A Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 7, Topic B Mission 7, Topic E Mission 8, Topic B
2.A.1.2	Represent and describe repeating patterns involving shapes in a variety of contexts.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
2.A.2.1	Use objects and number lines to represent number sentences.	Mission 7, Topic A Mission 7, Topic E Mission 7, Topic F
2.A.2.2	Generate models and situations to represent number sentences and vice versa.	Mission 1, Topic B Mission 2, Topic D Mission 7, Topic B
2.A.2.3	Apply the commutative property, identity property, and number sense to find values for unknowns that make addition and subtraction number sentences true or false.	Grade 1, Mission 1, Topic D Grade 1, Mission 1, Topic H Grade 1, Mission 2, Topic C
Geometry & Measurement		
2.GM.1.1	Recognize regular and irregular trapezoids and hexagons.	Mission 8, Topic A
2.GM.1.2	Describe, compare, and classify two-dimensional figures according to their geometric attributes.	Mission 8, Topic A Mission 8, Topic B

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.GM.1.3	Compose and decompose two-dimensional shapes using triangles, squares, hexagons, trapezoids, and rhombi.	Mission 8, Topic B
2.GM.1.4	Sort three-dimensional shapes based on attributes such as number of faces, vertices, and edges.	Grade 1, Mission 5, Topic A
2.GM.1.5	Recognize right angles and classify angles as smaller or larger than a right angle.	Mission 8, Topic A Grade 4, Mission 4, Topic A
2.GM.2.1	Explain the relationship between the size of the unit of measurement and the number of units needed to measure the length of an object.	Mission 2, Topic C Mission 7, Topic D
2.GM.2.2	Explain the relationship between length and the numbers on a ruler by using a ruler to measure lengths to the nearest whole unit.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 7, Topic C Mission 7, Topic D
2.GM.2.3	Explore how varying shapes and styles of containers can have the same capacity.	Grade K, Mission 3, Topic D
2.GM.3.1	Distinguish between a.m. and p.m.	Mission 8, Topic D
2.GM.3.2	Read and write time to the quarter hour on an analog and digital clock.	Mission 8, Topic D
Data & Probability		
2.D.1.1	Explain that the length of a bar in a bar graph and the number of objects in a pictograph represents the number of data points for a given category.	Mission 7, Topic A
2.D.1.2	Organize a collection of data with up to four categories using pictographs and bar graphs in intervals of 1s, 2s, 5s or 10s.	Mission 7, Topic A
2.D.1.3	Write and solve one-step word problems involving addition or subtraction using data represented within pictographs and bar graphs with intervals of one.	Mission 7, Topic A

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.D.1.4	Draw conclusions and make predictions from information in a pictograph and bar graph.	Mission 7, Topic A Mission 7, Topic E

3rd Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Numbers & Operations		
3.N.1.1	Read, write, discuss, and represent whole numbers up to 100,000. Representations should include but are not limited to numerals, words, pictures, number lines, and manipulatives (e.g., $350 = 3 \text{ hundreds, } 5 \text{ tens} = 35 \text{ tens} = 3 \text{ hundreds, } 4 \text{ tens, } 10 \text{ ones}$).	Grade 4, Mission 1, Topic A
3.N.1.2	Use place value to describe whole numbers between 1,000 and 100,000 in terms of ten thousands, thousands, hundreds, tens and ones, including written, standard, and expanded forms.	Grade 4, Mission 1, Topic A
3.N.1.3	Applying knowledge of place values, use mental strategies (no written computations) to find 100 more or 100 less than a given number, 1,000 more or 1,000 less than a given number, and 10,000 more or 10,000 less than a given number, up to a five-digit number.	Grade 4, Mission 1, Topic B Grade 2, Mission 5, Topic A
3.N.1.4	Use place value to compare and order whole numbers, up to 100,000, using comparative language, numbers, and symbols.	Grade 4, Mission 5, Topic B
3.N.1.5	Use place value understanding to round numbers to the nearest thousand, ten-thousand and hundred thousand.	Grade 4, Mission 1, Topic C
3.N.2.1	Represent multiplication facts by modeling a variety of approaches (e.g., manipulatives, repeated addition, equal-sized groups, arrays, area models, equal jumps on a number line, skip counting).	Mission 1, Topic A Mission 1, Topic C Mission 1, Topic E Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.N.2.2	Demonstrate fluency with multiplication facts using factors up to 10.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
		Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.N.2.3	Use strategies and algorithms based on knowledge of place value and equality to fluently add and subtract up to five-digit numbers (answer not to exceed 100,000).	Mission 2, Topic A Mission 2, Topic D Mission 2, Topic E
3.N.2.4	Recognize when to round numbers and apply understanding to estimate sums and differences to the nearest ten thousand, thousand, hundred, and ten.	Mission 2, Topic C Grade 4, Mission 1, Topic C
3.N.2.5	Use addition and subtraction to solve problems involving whole numbers. Use various strategies, including the relationship between addition and subtraction and the context of the problem to assess the reasonableness of results.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic D Mission 2, Topic E Mission 3, Topic E Mission 7, Topic A
3.N.2.6	Represent division facts and divisibility by modeling a variety of approaches (e.g., repeated subtraction, equal sharing, forming equal groups) to show the relationship between multiplication and division.	Mission 1, Topic B Mission 1, Topic D Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
3.N.2.7	Apply the relationship between multiplication and division to represent and solve problems.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
3.N.2.8	Use various strategies (e.g., base ten blocks, area models, arrays, repeated addition, algorithms) based	Mission 3, Topic F

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	on knowledge of place value, equality, and properties of addition and multiplication to multiply a two-digit factor by a one-digit factor.	
3.N.3.1	Read and write fractions with words and symbols using appropriate terminology (i.e., numerator and denominator).	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
3.N.3.2	Model fractions using length, set, and area for halves, thirds, fourths, sixths, and eighths.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C
3.N.3.3	Apply understanding of unit fractions and use this understanding to compose and decompose fractions related to the same whole.	Mission 5, Topic B Mission 5, Topic C
3.N.3.4	Use models and number lines to order and compare fractions that are related to the same whole.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic F
3.N.4.1	Use addition and subtraction to determine the value of a collection of coins up to one dollar using the cent symbol and in monetary transactions.	Grade 2, Mission 7, Topic B
3.N.4.2	Add and subtract a collection of bills up to twenty dollars using whole dollars in monetary transactions.	Grade 2, Mission 7, Topic B
Algebraic Reasoning & Algebra		
3.A.1.1	Create, describe, and extend patterns involving addition, subtraction, or multiplication to solve problems in a variety of contexts.	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F
3.A.1.2	Describe the rule (limited to a single operation) for a pattern from an input/output table or function machine involving addition, subtraction, or multiplication.	Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic F

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.A.1.3	Explore and develop visual representations of increasing and decreasing geometric patterns and construct the next steps.	Mission 7, Topic C
3.A.2.1	Use number sense with the properties of addition, subtraction, and multiplication, to find unknowns (represented by symbols) in one-step equations. Generate real-world situations to represent number sentences.	Mission 1, Topic B Mission 1, Topic C Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E Mission 7, Topic A
3.A.2.2	Identify, represent, and apply the number properties (commutative, identity, and associative properties of addition and multiplication) using models and manipulatives to solve problems.	Mission 1, Topic C Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic F
Geometry & Measurement		
3.GM.1.1	Sort three-dimensional shapes based on attributes.	Mission 7, Topic B
3.GM.1.2	Build a three-dimensional figure using unit cubes when shown a picture of a three-dimensional shape.	Grade 5, Mission 5, Topic A
3.GM.1.3	Classify angles within a polygon as acute, right, obtuse, and straight.	Grade 4, Mission 4, Topic A
3.GM.2.1	Find the perimeter of a polygon, given whole number lengths of the sides, using a variety of models.	Mission 7, Topic C Mission 7, Topic D Mission 7, Topic E
3.GM.2.2	Analyze why length and width are multiplied to find the area of a rectangle by decomposing the rectangle into one unit by one unit squares and viewing these as rows and columns to determine the area.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.GM.2.3	Count cubes systematically to identify the number of cubes needed to pack the whole or half of a three-dimensional structure.	Grade 5, Mission 5, Topic A
3.GM.2.4	Find the area of two-dimensional figures by counting the total number of same-size unit squares that fill the shape without gaps or overlaps.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic D
3.GM.2.5	Choose an appropriate measurement instrument and measure the length of objects to the nearest whole centimeter or whole meter.	Grade 2, Mission 2, Topic B Grade 2, Mission 2, Topic C
3.GM.2.6	Choose an appropriate measurement instrument and measure the length of objects to the nearest whole yard, whole foot, or half inch.	Mission 6, Topic B Grade 2, Mission 7, Topic D
3.GM.2.7.	Use an analog thermometer to determine temperature to the nearest degree in Fahrenheit and Celsius.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
3.GM.3.1	Read and write time to the nearest five-minute interval (analog and digital).	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
3.GM.3.2	Determine the solutions to problems involving addition and subtraction of time in intervals of five minutes, up to one hour, using pictorial models, number line diagrams, or other tools.	Mission 2, Topic A Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
Data & Probability		
3.D.1.1	Collect and organize a data set with multiple categories using a frequency table, line plot, pictograph, or bar graph with scaled intervals.	Mission 6, Topic A
3.D.1.2	Solve one- and two-step problems using categorical data represented with a frequency table, pictograph, or bar graph with scaled intervals.	Mission 6, Topic A

4th Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Numbers & Operations		
4.N.1.1	Read, write, discuss, and represent whole numbers up to 1,000,000. Representations may include numerals, words, pictures, number lines, and manipulatives.	Mission 1, Topic A
4.N.1.2	Use place value to describe whole numbers between 1,000 and 1,000,000 in terms of millions, hundred thousands, ten thousands, thousands, hundreds, tens, and ones with written, standard, and expanded forms.	Mission 1, Topic A
4.N.1.3	Applying knowledge of place value, use mental strategies (no written computations) to multiply or divide a number by 10, 100 and 1,000.	Mission 3, Topic B Mission 3, Topic G
4.N.1.4	Use place value to compare and order whole numbers up to 1,000,000, using comparative language, numbers, and symbols.	Mission 1, Topic B
4.N.2.1	Demonstrate fluency with multiplication and division facts with factors up to 12.	Grade 3, Mission 3, Topic A Grade 3, Mission 3, Topic B Grade 3, Mission 3, Topic C Grade 3, Mission 3, Topic D Grade 3, Mission 3, Topic E
4.N.2.2	Multiply 3-digit by 1-digit and 2-digit by 2-digit whole numbers, using various strategies, including but not limited to standard algorithms.	Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 3, Topic H
4.N.2.3	Estimate products of 3-digit by 1-digit and 2-digit by 2-digit whole number factors using a variety of strategies (e.g., rounding, front end estimation, adjusting, compatible numbers) to assess the reasonableness of results. Explore larger numbers using technology to investigate patterns.	Mission 3, Topic D
4.N.2.4	Apply and analyze models to solve multi-step problems requiring the use of addition, subtraction,	Mission 1, Topic D Mission 1, Topic E

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
	and multiplication of multi-digit whole numbers. Use various strategies, including the relationship between operations, the use of appropriate technology, and the context of the problem to assess the reasonableness of results.	Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C
4.N.2.5	Use strategies and algorithms (e.g., mental strategies, standard algorithms, partial quotients, repeated subtraction, the commutative, associative, and distributive properties) based on knowledge of place value, equality, and properties of operations to divide a 3-digit dividend by a 1-digit whole number divisor, with and without remainders.	Mission 3, Topic E Mission 3, Topic F Mission 3, Topic G
4.N.3.1	Represent and rename equivalent fractions using fraction models (e.g., parts of a set, area models, fraction strips, number lines).	Mission 5, Topic B Mission 5, Topic D Mission 5, Topic E Mission 6, Topic B
4.N.3.2	Use benchmark fractions (0, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, 1) to locate additional fractions with denominators up to twelfths on a number line.	Mission 5, Topic C Mission 5, Topic E
4.N.3.3	Use models to order and compare whole numbers and fractions less than and greater than one, using comparative language and symbols.	Mission 5, Topic C Mission 5, Topic E
4.N.3.4	Decompose a fraction into a sum of fractions with the same denominator in more than one way, using concrete and pictorial models and recording results with numerical representations (e.g., $\frac{3}{4} = \frac{1}{4} + \frac{1}{4} + \frac{1}{4}$ and $\frac{3}{4} = \frac{2}{4} + \frac{1}{4}$).	Mission 5, Topic A Mission 5, Topic E
4.N.3.5	Use models to add and subtract fractions with like denominators.	Mission 5, Topic C Mission 5, Topic E
4.N.3.6	Represent tenths and hundredths with concrete and pictorial models, making connections between fractions and decimals.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic D Mission 6, Topic E

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.N.3.7	Read and write decimals in standard, word, and expanded form up to at least the hundredths place in a variety of contexts, including money.	Mission 6, Topic B Mission 6, Topic C Mission 6, Topic D Mission 6, Topic E
4.N.3.8	Compare and order decimals and whole numbers using place value and various models including but not limited to grids, number lines, and base 10 blocks.	Mission 5, Topic C Mission 5, Topic E Mission 6, Topic C
4.N.3.9	Compare and order benchmark fractions (0 , $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, 1) and decimals (0 , 0.25 , 0.50 , 0.75 , 1.00) in a variety of representations.	Mission 5, Topic C Mission 6, Topic C
4.N.4.1	Select the fewest number of coins for a given amount of money up to one dollar.	Grade 2, Mission 7, Topic B
4.N.4.2	Given a total cost (dollars and coins up to twenty dollars) and amount paid (dollars and coins up to twenty dollars), find the change required in a variety of ways.	Mission 6, Topic E
Algebraic Reasoning & Algebra		
4.A.1.1	Create an input/output chart or table to represent or extend a numerical pattern.	Mission 5, Topic H
4.A.1.2	Describe the single operation rule for a pattern from an input/output table or function machine involving any operation of a whole number.	Mission 5, Topic H
4.A.1.3	Construct models to show growth patterns involving geometric shapes and define the single operation rule of the pattern.	Grade 5, Mission 6, Topic F
4.A.2.1	Use the relationships between multiplication and division with the properties of multiplication to solve problems and find values for variables that make number sentences true.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 5, Topic G Mission 7, Topic A Mission 7, Topic B

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.A.2.2	Solve for a variable in an equation involving addition, subtraction, multiplication, or division with whole numbers. Analyze models to represent number sentences and vice versa.	Mission 1, Topic D Mission 1, Topic E Mission 1, Topic F Mission 3, Topic A Mission 3, Topic D Mission 3, Topic E Mission 3, Topic G Mission 3, Topic H Mission 7, Topic B Mission 7, Topic C
4.A.2.3	Determine the unknown addend or factor in equivalent and non-equivalent expressions (e.g., $5 + 6 = 4 + \square$, $8 < 3 \square$).	<i>Zearn Math includes digital activities that require students to find unknown addends or factors; however, the Zearn Math digital activities do not include non-equivalent expressions. It is recommended that teachers use supplemental materials to have students find unknown addends or factors in non-equivalent expressions.</i>
Geometry & Measurement		
4.GM.1.1	Identify points, lines, line segments, rays, angles, endpoints, and parallel and perpendicular lines in various models.	Mission 4, Topic A Mission 4, Topic D
4.GM.1.2	Describe, classify, and construct quadrilaterals, including squares, rectangles, trapezoids, rhombuses, parallelograms, and kites. Recognize quadrilaterals in various models.	Mission 4, Topic D
4.GM.1.3	Given two three-dimensional shapes, identify each shape. Compare and contrast their similarities and differences based on their attributes.	Grade 6, Mission 1, Topic E
4.GM.2.1	Measure angles in geometric figures and real-world objects with a protractor or angle ruler.	Mission 4, Topic B
4.GM.2.2	Find the area of polygons by determining if they can be decomposed into rectangles.	Grade 3, Mission 4, Topic D

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.GM.2.3	Develop the concept that the volume of rectangular prisms with whole-number edge lengths can be found by counting the total number of same-sized unit cubes that fill a shape without gaps or overlaps. Use a variety of tools and create models to determine the volume using appropriate measurements (e.g., cm ³).	Grade 5, Mission 5, Topic A
4.GM.2.4	Choose an appropriate instrument to measure the length of an object to the nearest whole centimeter or quarter inch.	Grade 2, Mission 7, Topic C Grade 2, Mission 7, Topic D
4.GM.2.5	Recognize and use the relationship between inches, feet, and yards to measure and compare objects.	Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
4.GM.2.6	Recognize and use the relationship between millimeters, centimeters, and meters to measure and compare objects.	Mission 2, Topic A Mission 2, Topic B
4.GM.2.7	Determine and justify the best use of customary and metric measurements in a variety of situations (liquid volumes, mass vs. weight, temperatures above 0 (zero) degrees, and length).	Mission 2, Topic A Mission 2, Topic B Mission 6, Topic C Mission 6, Topic E Mission 7, Topic A Mission 7, Topic B Mission 7, Topic C
4.GM.3.1	Determine elapsed time.	Grade 3, Mission 2, Topic A
4.GM.3.2	Convert one measure of time to another including seconds to minutes, minutes to hours, hours to days, and vice versa, using various models	Mission 7, Topic A
Data & Probability		
4.MD.1	1 Create and organize data on a frequency table or line plot marked with whole numbers and fractions using appropriate titles, labels, and units.	Mission 5, Topic E Mission 5, Topic G

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.MD.2	Organize data sets to create tables, bar graphs, timelines, and Venn diagrams. The data may include benchmark fractions or decimals ($\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$, $\frac{3}{4}$, 0.25, 0.50, 0.75).	Mission 5, Topic E Mission 5, Topic G
4.MD.3	Solve one- and two-step problems by analyzing data in whole number, decimal, or fraction form in a frequency table and line plot.	Mission 5, Topic E Mission 5, Topic G

5th Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Numbers & Operations		
5.N.1.1	Represent decimal fractions (e.g. $\frac{1}{10}$, $\frac{1}{100}$) using a variety of models (e.g., 10 by 10 grids, base-ten blocks, meter stick) and show the rational number relationships among fractions, decimals and whole numbers.	Grade 4, Mission 6, Topic A Grade 4, Mission 6, Topic B Grade 4, Mission 6, Topic C
5.N.1.2	Read, write, and represent decimals using place value to describe decimal numbers including fractional numbers as small as thousandths and whole numbers up to seven digits.	Mission 1, Topic A Mission 1, Topic B
5.N.1.3	Compare and order decimals and fractions, including mixed numbers and fractions less than one, and locate on a number line.	Mission 1, Topic B
5.N.1.4	Recognize and generate equivalent terminating decimals, fractions, mixed numbers, and fractions in various models	Mission 3, Topic A Mission 4, Topic D Grade 4, Mission 5, Topic B Grade 4, Mission 5, Topic D Grade 4, Mission 5, Topic E Grade 4, Mission 6, Topic B
5.N.2.1	Estimate solutions to division problems to assess the reasonableness of results.	Mission 2, Topic E Mission 2, Topic F Mission 2, Topic H
5.N.2.2	Divide multi-digit numbers, by one- and two-digit divisors, based on knowledge of place value, including but not limited to standard algorithms.	Mission 2, Topic E Mission 2, Topic F Mission 2, Topic H
5.N.2.3	Recognize that remainders can be represented in a variety of ways, including a whole number, fraction, or decimal. Determine the most meaningful form of a remainder based on the context of the problem.	Mission 4, Topic B
5.N.2.4	Construct models to solve multi-digit whole number problems requiring addition, subtraction,	Mission 2, Topic B Mission 2, Topic C

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	multiplication, and division using various representations, including the inverse relationships between operations, the use of technology, and the context of the problem to assess the reasonableness of results.	Mission 2, Topic G
5.N.3.1	Estimate sums and differences of fractions with like and unlike denominators, mixed numbers, and decimals to assess the reasonableness of the results.	Mission 1, Topic D Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic D
5.N.3.2	Illustrate addition and subtraction of fractions with like and unlike denominators, mixed numbers, and decimals using a variety of mathematical models (e.g., fraction strips, area models, number lines, fraction rods).	Mission 1, Topic D Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D
5.N.3.3	Add and subtract fractions with like and unlike denominators, mixed numbers, and decimals, involving money, measurement, geometry, and data. Use various models and efficient strategies, including but not limited to standard algorithms.	Mission 1, Topic D Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C Mission 3, Topic D Mission 4, Topic D
5.N.3.4	Apply mental math and knowledge of place value (no written computations) to find 0.1 more or 0.1 less than a number, 0.01 more or 0.01 less than a number, and 0.001 more or 0.001 less than a number.	Mission 1, Topic D
Algebraic Reasoning & Algebra		
5.A.1.1	Use tables and rules with up to two operations to describe patterns of change and make predictions and generalizations about various mathematical situations.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
5.A.1.1	Use a rule or table to represent ordered pairs of whole numbers and graph these ordered pairs on a coordinate plane, identifying the origin and axes in relation to the coordinates.	Mission 6, Topic C Mission 6, Topic D

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.A.2.1	Generate equivalent numerical expressions and solve problems using number sense involving whole numbers by applying the commutative property, associative property, distributive property, and order of operations (excluding exponents).	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 4, Topic D Mission 4, Topic G Mission 4, Topic H
5.A.2.2	Determine whether an equation or inequality involving a variable is true or false for a given value of the variable.	Grade 6, Mission 6, Topic A Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C
5.A.2.3	Evaluate expressions involving variables when values for the variables are given	Mission 1, Topic B Mission 1, Topic C Mission 6, Topic B Mission 6, Topic C
Geometry & Measurement		
5.GM.1.1	Describe, identify, classify, and construct triangles (equilateral, right, scalene, isosceles) by their attributes using various mathematical models.	Grade 4, Mission 4, Topic D
5.GM.1.2	Describe, identify, and classify three-dimensional figures (cubes, rectangular prisms, and pyramids) and their attributes (number of edges, faces, vertices, shapes of faces), given various mathematical models.	Mission 5, Topic A Mission 5, Topic B
5.GM.1.3	Recognize and draw a net for a three-dimensional figure (cube, rectangular prism, pyramid).	Grade 6, Mission 1, Topic E Grade 6, Mission 1, Topic F Grade 6, Mission 1, Topic G
5.GM.2.1	Determine the volume of rectangular prisms by the number of unit cubes (n) used to construct the shape and by the product of the dimensions of the prism $a \cdot b \cdot c = n$. Understand rectangular prisms of different dimensions (p , q , and r) can have the same volume if $a \cdot b \cdot c = p \cdot q \cdot r = n$.	Mission 5, Topic B
5.GM.2.2	Estimate the perimeter of polygons and create arguments for reasonable perimeter values of shapes that may include curves.	Grade 4, Mission 3, Topic A Grade 4, Mission 3, Topic H

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.GM.3.1	Measure and compare angles according to size using various tools.	Grade 4, Mission 4, Topic B
5.GM.3.2	Measure the length of an object to the nearest whole centimeter or up to 1/16 inch using an appropriate instrument	Mission 4, Topic A
5.GM.3.3	Apply the relationship between inches, feet, and yards to measure, convert, and compare objects to solve problems.	Mission 4, Topic A Grade 2, Mission 7, Topic D
5.GM.3.4	Apply the relationship between millimeters, centimeters, and meters to measure, convert, and compare objects to solve problems.	Mission 1, Topic A Mission 2, Topic D Mission 4, Topic C Mission 4, Topic E
5.GM.3.5	Estimate lengths and geometric measurements to the nearest whole unit, using benchmarks in customary and metric measurement systems.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
Data & Probability		
5.D.1.1	Find the measures of central tendency (i.e., mean, median, mode) and range of a set of data. Understand that the mean is a “leveling out” or central balance point of the data.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
5.D.1.2	Create and analyze line and double-bar graphs with increments of whole numbers, fractions, and decimals.	Mission 6, Topic C

6th Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Numbers & Operations		
6.N.1.1	Use manipulatives and models (e.g., number lines) to determine positive and negative numbers and their contexts, identify opposites, and explain the meaning of 0 (zero) in a variety of situations.	Mission 7, Topic A Mission 7, Topic C
6.N.1.2	Compare and order positive rational numbers, represented in various forms, or integers using the symbols "<", ">", and "=".	Grade 5, Mission 1, Topic B
6.N.1.3	Explain that a percent represents parts “out of 100” and ratios “to 100.”	Mission 3, Topic D
6.N.1.4	Determine equivalencies among fractions, mixed numbers, decimals, and percents.	Mission 3, Topic D
6.N.2.1	Estimate solutions for integer addition and subtraction of problems in order to assess the reasonableness of results.	Grade 7, Mission 5, Topic A Grade 7, Mission 5, Topic C
6.N.2.2	Illustrate addition and subtraction of integers using a variety of representations.	Grade 7, Mission 5, Topic A Grade 7, Mission 5, Topic C
6.N.2.3	Add and subtract integers in a variety of situations; use efficient and generalizable procedures including but not limited to standard algorithms.	Grade 7, Mission 5, Topic A Grade 7, Mission 5, Topic C
6.N.2.4	Identify and represent patterns with whole-number exponents and perfect squares. Evaluate powers with whole-number bases and exponents.	Mission C, Topic C
6.N.2.5	Factor whole numbers and express prime and composite numbers as a product of prime factors with exponents.	Mission 7, Topic D
6.N.2.6	Determine the greatest common factors and least common multiples. Use common factors and	Mission 7, Topic D

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	multiplies to calculate with fractions, find equivalent fractions, and express the sum of two-digit numbers with a common factor using the distributive property.	
6.N.3.1	Identify and use ratios to compare and relate quantities in multiple ways. Recognize that multiplicative comparison and additive comparison are different.	Mission 2, Topic A Mission 2, Topic B
6.N.3.2	Determine the unit rate for ratios.	Mission 2, Topic C Mission 3, Topic A Mission 3, Topic C
6.N.3.3	Apply the relationship between ratios, equivalent fractions, unit rates, and percents to solve problems in various contexts.	Mission 2, Topic C Mission 3, Topic C Mission 3, Topic D Mission 3, Topic E
6.N.4.1	Estimate solutions to problems with whole numbers, decimals, fractions, and mixed numbers, and use the estimates to assess the reasonableness of results in the context of the problem.	Grade 7, Mission 3, Topic C Grade 7, Mission 5, Topic C Grade 7, Mission 5, Topic F Grade 7, Mission 6, Topic A Grade 7, Mission 6, Topic B
6.N.4.2	Illustrate multiplication and division of fractions and decimals to show connections to fractions, whole number multiplication, and inverse relationships.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
6.N.4.3	Multiply and divide fractions and decimals using efficient and generalizable procedures.	Mission 4, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D Mission 4, Topic E Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.N.4.4	Use mathematical modeling to solve and interpret problems including money, measurement, geometry, and data requiring arithmetic with decimals, fractions and mixed numbers.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
Algebraic Reasoning & Algebra		
6.A.1.1	Plot integer- and rational-valued (limited to halves and fourths) ordered-pairs as coordinates in all four quadrants and recognize the reflective relationships among coordinates that differ only by their signs.	Mission 7, Topic A Mission 7, Topic C
6.A.1.2	Represent relationships between two varying positive quantities involving no more than two operations with rules, graphs, and tables; translate between any two of these representations.	Mission 6, Topic D
6.A.1.3	Use and evaluate variables in expressions, equations, and inequalities that arise from various contexts, including determining when or if, for a given value of the variable, an equation or inequality involving a variable is true or false.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.A.2.1	Generate equivalent expressions and evaluate expressions involving positive rational numbers by applying the commutative, associative, and distributive properties and order of operations to model and solve mathematical problems.	Mission 6, Topic B Mission 6, Topic C
6.A.3.1	Model mathematical situations using expressions, equations and inequalities involving variables and rational numbers.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C Mission 7, Topic B
6.A.3.2	Use number sense and properties of operations and equality to model and solve mathematical problems involving equations in the form $x + p = q$ and $px = q$, where p and q are nonnegative rational numbers. Graph the solution on a number line, interpret the solution in the original context, and assess the reasonableness of the solution.	Mission 6, Topic A

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Geometry & Measurement		
6.GM.1.1	Predict, describe, and apply translations (slides), reflections (flips), and rotations (turns) to a two-dimensional figure.	Grade 8, Mission 1, Topic A Grade 8, Mission 1, Topic B
6.GM.1.2	Recognize that translations, reflections, and rotations preserve congruence and use them to show that two figures are congruent.	Grade 8, Mission 1, Topic C Grade 8, Mission 1, Topic D
6.GM.1.3	Identify and describe the line(s) of symmetry in two-dimensional shapes.	Grade 4, Mission 4, Topic D
6.GM.2.1	Develop and use formulas for the area of squares and parallelograms using a variety of methods including but not limited to the standard algorithms and finding unknown measures.	Mission 6, Topic A Mission 6, Topic B
6.GM.2.2	Develop and use formulas to determine the area of triangles and find unknown measures.	Mission 6, Topic C
6.GM.2.3	Find the area of right triangles, other triangles, special quadrilaterals, and polygons that can be decomposed into triangles and other shapes.	Mission 6, Topic A Mission 6, Topic B Mission 6, Topic C
6.GM.3.1	Solve problems using the relationships between the angles (vertical, complementary, and supplementary) formed by intersecting lines.	Grade 7, Mission 7, Topic A
6.GM.3.2	Develop and use the fact that the sum of the interior angles of a triangle is 180° to determine missing angle measures in a triangle.	Grade 8, Mission 1, Topic D
6.GM.4.1	Estimate weights and capacities using benchmarks in customary and metric measurement systems with appropriate units.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
6.GM.4.2	Solve problems that require the conversion of lengths within the same measurement systems using appropriate units.	Mission 3, Topic B
Data & Probability		

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.D.1.1	Interpret the mean, median, and mode for a set of data.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
6.D.1.2	Explain and justify which measure of center (mean, median, or mode) would provide the most descriptive information for a given set of data.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
6.D.2.1	Represent possible outcomes using a probability continuum from impossible to certain.	Grade 7, Mission 8, Topic A
6.D.2.2	Determine the sample space for a given experiment and determine which members of the sample space are related to certain events. Sample space may be determined by the use of tree diagrams, tables or pictorial representations.	Grade 7, Mission 8, Topic A Grade 7, Mission 8, Topic B
6.D.2.3	Demonstrate simple experiments in which the probabilities are known and compare the resulting relative frequencies with the known probabilities, recognizing that there may be differences between the two results.	Grade 7, Mission 8, Topic A Grade 7, Mission 8, Topic B Grade 7, Mission 8, Topic E

7th Grade

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Numbers & Operations		
7.N.1.1	Compare and order rational numbers expressed in various forms using the symbols "<", ">", and "=".	Grade 6, Mission 7, Topic A Grade 6, Mission 7, Topic B
7.N.1.2	Recognize and generate equivalent representations of rational numbers, including equivalent fractions.	Mission 6, Topic D
7.N.1.3	Explain the relationship between the absolute value of a rational number and the distance of that number from zero on a number line. Use the symbol for absolute value. Apply the concept of absolute value to model and solve problems.	Grade 6, Mission 7, Topic A
7.N.2.1	Estimate solutions to multiplication and division of integers in order to assess the reasonableness of results.	Mission 5, Topic A Mission 5, Topic C
7.N.2.2	Illustrate multiplication and division of integers using a variety of representations.	Mission 5, Topic A Mission 5, Topic C
7.N.2.3	Multiply and divide integers in a variety of situations; use efficient and generalizable procedures, including standard algorithms.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
7.N.2.4	Raise rational numbers (integers, fractions, and decimals) to positive integer exponents.	Grade 6, Mission 6, Topic C
7.N.2.5	Model and solve problems using rational numbers involving addition, subtraction, multiplication, division, and positive integer exponents.	Grade 6, Mission 6, Topic C Mission 5, Topic B Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Algebraic Reasoning & Algebra		

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.A.1.1	Identify a relationship between two varying quantities, x and y , as proportional if it can be expressed in the form $y/z = k$ or $y = kx$; distinguish proportional relationships from non-proportional relationships.	Mission 2, Topic B Mission 2, Topic C
7.A.1.2	Recognize that the graph of a proportional relationship is a line through the origin and the coordinate $(1, r)$, where r is the slope and the unit rate (constant of proportionality, k).	Mission 2, Topic D
7.A.2.1	Represent proportional relationships with tables, verbal descriptions, symbols, and graphs; translate from one representation to another. Determine and compare the unit rate (constant of proportionality, slope, or rate of change) given any of these representations.	Mission 2, Topic A Mission 2, Topic B Mission 2, Topic C Mission 2, Topic D Mission 2, Topic E
7.A.2.2	Solve multi-step problems with proportional relationships (e.g., distance-time, percent increase or decrease, discounts, tips, unit pricing, mixtures and concentrations, similar figures, other mathematical situations).	Mission 3, Topic A Mission 4, Topic B Mission 4, Topic C Mission 4, Topic D
7.A.2.3	Use proportional reasoning to solve problems involving ratios.	Mission 2, Topic C Mission 4, Topic A
7.A.2.4	Use proportional reasoning to assess the reasonableness of solutions.	Mission 6, Topic B
7.A.3.1	Write and solve problems leading to linear equations with one variable in the form $px + q = r$ and $p(x + q) = r$, where p , q , and r are rational numbers.	Mission 6, Topic A Mission 6, Topic B
7.A.3.2	Represent, write, solve, and graph problems leading to linear inequalities with one variable in the form $x + p > q$ and $x + p < q$, where p , and q are nonnegative rational numbers.	Mission 6, Topic C
7.A.4.1	Use properties of operations (associative, commutative, and distributive) to generate equivalent numerical and algebraic expressions containing	Mission 6, Topic D

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
	rational numbers, grouping symbols and whole number exponents.	
7.A.4.2	Evaluate numerical expressions using calculators and other technologies and justify solutions using order of operations and grouping symbols.	Grade 7, Mission 5, Topic C Grade 7, Mission 5, Topic D Grade 7, Mission 6, Topic D <i>These topics focus on evaluating numerical expressions using order of operations and grouping symbols, not technology. It is recommended that teachers use supplemental materials to address the depth of this standard.</i>
Geometry & Measurement		
7.GM.1.1	Recognize that the surface area of a rectangular prism can be found by finding the area of each component of the net of that figure. Know that rectangular prisms of different dimensions can have the same surface area.	Grade 6, Mission 1, Topic E Grade 6, Mission 1, Topic F Grade 6, Mission 1, Topic G
7.GM.1.2	Using a variety of tools and strategies, develop the concept that surface area of a rectangular prism can be found by wrapping the figure with same-sized square units without gaps or overlap. Use appropriate measurements (e.g., cm^2).	Grade 6, Mission 1, Topic E Grade 6, Mission 1, Topic F Grade 6, Mission 1, Topic G
7.GM.1.3	Using a variety of tools and strategies, develop the concept that the volume of rectangular prisms can be found by counting the total number of same-sized unit cubes that fill a shape without gaps or overlaps. Use appropriate measurements (e.g., cm^3).	Grade 5, Mission 5, Topic A Grade 5, Mission 5, Topic B
7.GM.2.1	Develop and use the formula to determine the area of a trapezoid.	Grade 6, Mission 1, Topic A Grade 6, Mission 1, Topic B <i>These topics focus on finding the area of composite figures. It is recommended that teachers use supplemental materials to address the depth of this standard.</i>

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.GM.2.2	Find the area and perimeter of composite figures.	Grade 6, Mission 1, Topic A Grade 6, Mission 1, Topic B
7.GM.3.1	Solve problems that require the conversion of weights and capacities within the same measurement systems using appropriate units.	Grade 8, Mission 1, Topic C Grade 8, Mission 1, Topic D
7.GM.3.2	Demonstrate an understanding of the proportional relationship between the diameter and circumference of a circle and that the unit rate (constant of proportionality) is pi (π) and can be approximated by rational numbers such as $\frac{22}{7}$ and 3.14.	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.GM.3.3	Calculate the circumference and area of circles to solve problems in various contexts, in terms of pi (π) and using approximations for pi (π).	Mission 3, Topic A Mission 3, Topic B Mission 3, Topic C
7.GM.4.1	Describe the properties of similarity, compare geometric figures for similarity, and determine scale factors resulting from dilations.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C
7.GM.4.2	Apply proportions, ratios, and scale factors to solve problems involving scale drawings and to determine side lengths and areas of similar triangles and rectangles.	Mission 1, Topic A Mission 1, Topic B Mission 1, Topic C
7.GM.4.3	Graph and describe translations (with directional and algebraic instructions), reflections across the x- and y-axes, and rotations in 90° increments about the origin of figures on a coordinate plane, and determine the coordinates of the vertices of a figure after a transformation.	Grade 8, Mission 1, Topic A
Data & Probability		
7.D.1.1	Design simple experiments, collect data, and calculate measures of center (mean, median, and mode) and spread (range and interquartile range). Use these quantities to draw conclusions about the data collected and make predictions.	Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.D.1.2	Use reasoning with proportions to display and interpret data in circle graphs (pie charts) and histograms.	Grade 6, Mission 8, Topic B
7.D.1.3	Use technology to create and analyze box plots.	Grade 6, Mission 8, Topic D
7.D.2.1	Determine the theoretical probability of an event using the ratio between the size of the event and the size of the sample space; represent probabilities as percents, fractions and decimals between 0 and 1.	Mission 8, Topic A
7.D.2.2	Calculate probability as a fraction of sample space or as a fraction of area. Express probabilities as percents, decimals and fractions.	Mission 8, Topic A
7.D.2.3	Use proportional reasoning to draw conclusions about and predict relative frequencies of outcomes based on theoretical probabilities.	Mission 8, Topic A

Pre-Algebra

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Pre-Algebra Standards		Lessons
Numbers & Operations		
PA.N.1.1	Develop and apply the properties of integer exponents, including $a^0 = 1$ (with $a \neq 0$), to generate equivalent numerical and algebraic expressions.	Mission 7, Topic B Mission 7, Topic C
PA.N.1.2	Express and compare approximations of very large and very small numbers using scientific notation.	Mission 7, Topic C Mission 7, Topic D
PA.N.1.3	Multiply and divide numbers expressed in scientific notation and express the answer in scientific notation.	Mission 7, Topic C Mission 7, Topic D
PA.N.1.4	Compare and order real numbers; locate real numbers on a number line. Identify the square roots of perfect squares to 400 or, if it is not a perfect square root, locate it as an irrational number between two consecutive positive integers.	Mission 8, Topic A Mission 8, Topic B Mission 8, Topic C Mission 8, Topic D Mission 8, Topic E
Algebraic Reasoning & Algebra		
PA.A.1.1	Recognize that a function is a relationship between an independent variable and a dependent variable in which the value of the independent variable determines the value of the dependent variable.	Mission 5, Topic A Mission 5, Topic B Mission 5, Topic E
PA.A.1.2	Use linear functions to represent and model mathematical situations.	Mission 5, Topic C Mission 5, Topic D Mission 5, Topic E
PA.A.1.3	Identify a function as linear if it can be expressed in the form $y=mx + b$ or if its graph is a non-vertical straight line.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
PA.A.2.1	Represent linear functions with tables, verbal descriptions, symbols, and graphs; translate from one representation to another.	Mission 5, Topic B Mission 5, Topic C

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Pre-Algebra Standards		Lessons
PA.A.2.2	Identify, describe, and analyze linear relationships between two variables.	Mission 5, Topic B Mission 5, Topic C Mission 5, Topic E
PA.A.2.3	Identify graphical properties of linear functions, including slope and intercepts. Know that the slope equals the rate of change, and that the y-intercept is zero when the function represents a proportional relationship.	Mission 2, Topic C Mission 3, Topic B Mission 3, Topic C Mission 3, Topic E
PA.A.2.4	Predict the effect on the graph of a linear function when the slope or y-intercept changes. Use appropriate tools to examine these effects.	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
PA.A.2.5	Solve problems involving linear functions and interpret results in the original context.	Grade 8, Mission 3, Topic B Grade 8, Mission 5, Topic B Grade 8, Mission 5, Topic C Grade 8, Mission 6, Topic B
PA.A.3.1	Use substitution to simplify and evaluate algebraic expressions.	Grade 6, Mission 6, Topic B Grade 6, Mission 6, Topic C
PA.A.3.2	Justify steps in generating equivalent expressions by combining like terms and using order of operations (to include grouping symbols). Identify the properties used, including the properties of operations (associative, commutative, and distributive).	Grade 7, Mission 6, Topic D
PA.A.4.1	Solve mathematical problems using linear equations with one variable where there could be one, infinitely many, or no solutions. Represent situations using linear equations and interpret solutions in the original context.	Mission 4, Topic B
PA.A.4.2	Represent, write, solve, and graph problems leading to linear inequalities with one variable in the form $px + q > r$ and $px + q < r$, where p, q, and r are rational numbers.	Grade 7, Mission 6, Topic C
PA.A.4.3	Represent real-world situations using equations and inequalities involving one variable.	Grade 7, Mission 6, Topic A Grade 7, Mission 6, Topic D Grade 7, Mission 6, Topic C Mission 4, Topic B

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Pre-Algebra Standards		Lessons
Geometry & Measurement		
PA.GM.1.1	Justify the Pythagorean theorem using measurements, diagrams, or dynamic software to solve problems in two dimensions involving right triangles.	Mission 8, Topic C
PA.GM.1.2	Use the Pythagorean theorem to find the distance between any two points in a coordinate plane.	Mission 8, Topic C
PA.GM.2.1	Calculate the surface area of a rectangular prism using decomposition or nets. Use appropriate units (e.g., cm^2).	Grade 6, Mission 1, Topic E
PA.GM.2.2	Calculate the surface area of a cylinder, in terms of π and using approximations for π (π), using decomposition or nets. Use appropriate units (e.g., cm^2).	<i>It is recommended that teachers use supplemental materials to fully address this standard.</i>
PA.GM.2.3	Justify why base area (B) and height (h) in the formula $V=Bh$ are multiplied to find the volume of a rectangular prism. Use appropriate units (e.g., cm^3).	Grade 5, Mission 5, Topic B
PA.GM.2.4	Develop and use the formulas $V = (\pi r)^2 h$ and $V = Bh$ to determine the volume of right cylinders, in terms of π and using approximations for π (π). Justify why base area (B) and height (h) are multiplied to find the volume of a right cylinder. Use appropriate units (e.g., cm^3).	Mission 5, Topic D Mission 5, Topic E Mission 5, Topic F
Data & Probability		
PA.D.1.1	Describe the impact that inserting or deleting a data point has on the mean and the median of a data set. Create data displays using technology to examine this impact.	Grade 6, Mission 8, Topic A Grade 6, Mission 8, Topic B Grade 6, Mission 8, Topic C Grade 6, Mission 8, Topic D Grade 6, Mission 8, Topic E
PA.D.1.2	Explain how outliers affect measures of center and spread.	Grade 6, Mission 8, Topic B Grade 6, Mission 8, Topic C Grade 6, Mission 8, Topic D Grade 7, Mission 8, Topic D Grade 7, Mission 8, Topic E

OKLAHOMA'S ACADEMIC STANDARDS FOR MATHEMATICS		ZEARN MATH
Pre-Algebra Standards		Lessons
PA.D.1.3	Collect, display, and interpret data using scatter plots. Use the shape of the scatter plot to find the informal line of best fit, make statements about the average rate of change, and make predictions about values not in the original data set. Use appropriate titles, labels, and units.	Mission 6, Topic A Mission 6, Topic B
PA.D.2.1	Calculate experimental probabilities and represent them as percents, fractions, and decimals between 0 and 1. Use experimental probabilities to predict relative frequencies when actual probabilities are unknown.	Grade 7, Mission 8, Topic A
PA.D.2.2	Determine how samples are chosen (randomness) to draw and support conclusions about generalizing a sample to a population, including identifying limitations and biases.	Grade 7, Mission 8, Topic C Grade 7, Mission 8, Topic D Grade 7, Mission 8, Topic E
PA.D.2.3	Define, compare, and contrast the probabilities of dependent and independent events.	Grade 7, Mission 8, Topic B <i>This topic focuses on defining and determining the probability of independent events. It is recommended that teachers use supplemental materials to address the depth of this standard.</i>